Performance benchmarking of HRNet for object detection in variable lighting conditions

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Abstract

Detecting and segmenting objects under variable illumination remains challenging for computer vision models. This project evaluates illumination sensitivity by grouping outdoor webcam images into morning, afternoon, evening, and night categories using timestamp metadata, then running a pre-trained High-Resolution Network (HRNet) semantic segmentation model. Because ground-truth segmentation masks are unavailable, proxy metrics are reported: average confidence and the average number of detected object classes (excluding background). Low-light enhancement and denoising are applied to night images, followed by re-segmentation and re-evaluation.

1 Introduction

Object detection and semantic segmentation are core tasks in computer vision with applications in surveillance, autonomous navigation, and scene understanding. Performance often degrades in low-light environments due to reduced contrast and increased noise. This report quantifies segmentation behavior under different lighting conditions by stratifying images by time-of-day and comparing model outputs.

2 Dataset

The Kaggle Day-night Dataset is used:

https://www.kaggle.com/datasets/stevemark/daynight-dataset

The dataset contains outdoor webcam imagery with associated timestamps, enabling grouping by time-of-day.

3 Methodology

3.1 Time-of-day categorization

Images are assigned to one of four time categories using the hour extracted from timestamp metadata:

- Morning: 05:00–11:59
- Afternoon: 12:00–16:59

- Evening: 17:00–20:59
- Night: 21:00–04:59

3.2 Segmentation model

A pre-trained HRNet semantic segmentation model from TensorFlow Hub is used: `google/HRNet/camvid-hrnetv2` Inference is performed without fine-tuning because pixel-level ground truth masks are unavailable.

3.3 Proxy performance metrics

For each image, the following quantities are computed:

1. **Confidence**: mean of the maximum predicted class probability at each pixel.
2. **Number of objects detected**: the count of unique predicted classes in the segmentation mask excluding background (interpreted as detected object categories).

Category-level averages are reported for morning, afternoon, evening, and night.

3.4 Night image enhancements

Night images are additionally processed with:

- **Low-light enhancement**: histogram equalization on the V channel in HSV color space.
- **Denoising**: non-local means denoising (`fastNlMeansDenoisingColored`).

Enhanced night images are re-segmented with the same HRNet model and re-evaluated using the same proxy metrics.

4 Results

4.1 Performance analysis by time of day

Table 1 reports baseline segmentation metrics by time-of-day.

Table 1: Performance Analysis by Time of Day (baseline HRNet inference).

Time of Day	Average Confidence	Average Objects Detected
Morning	0.9465	7.50
Afternoon	0.8881	8.00
Evening	0.9049	7.50
Night	0.8210	6.00

4.2 Night performance after enhancement and denoising

After applying low-light enhancement and denoising to night images, the following averages are obtained:

- Average Confidence: 0.8308
- Average Objects Detected: 8.00

5 Results Figures

5.1 Color legend

Figure 1: Color legend.

5.2 Segmentation Results with Overlay (Morning)

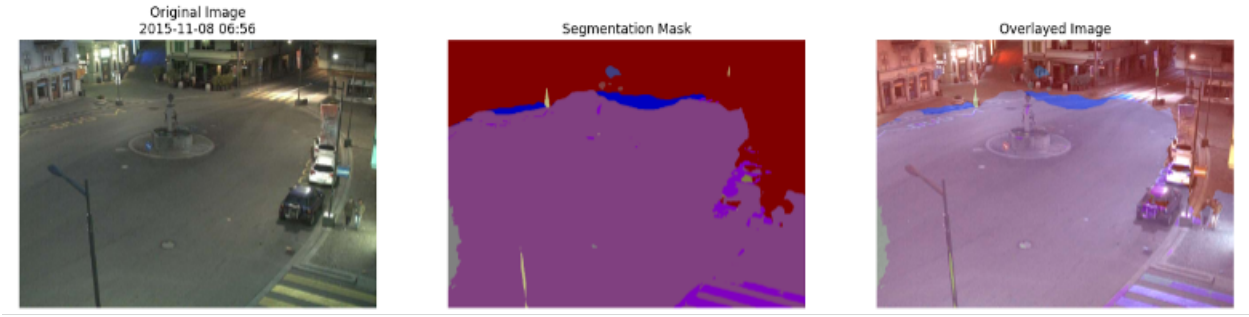

Figure 2: Segmentation Results with Overlay (Morning).

5.3 Segmentation Results with Overlay (Afternoon)

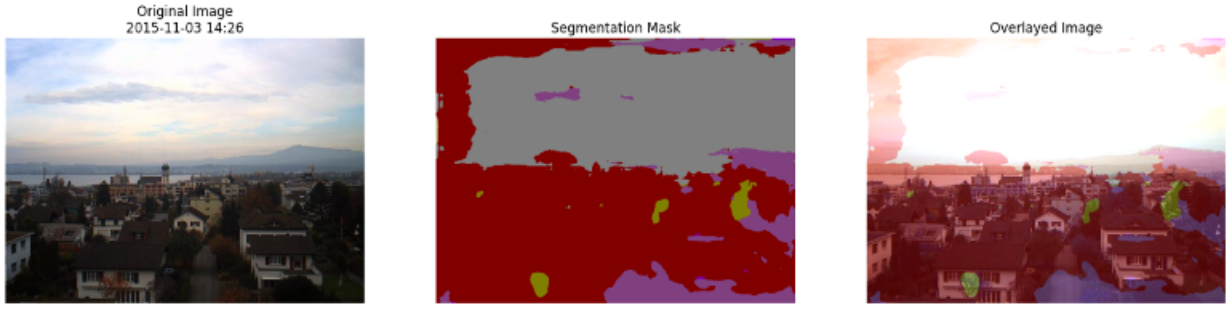

Figure 3: Segmentation Results with Overlay (Afternoon).

5.4 Segmentation Results with Overlay (Evening)

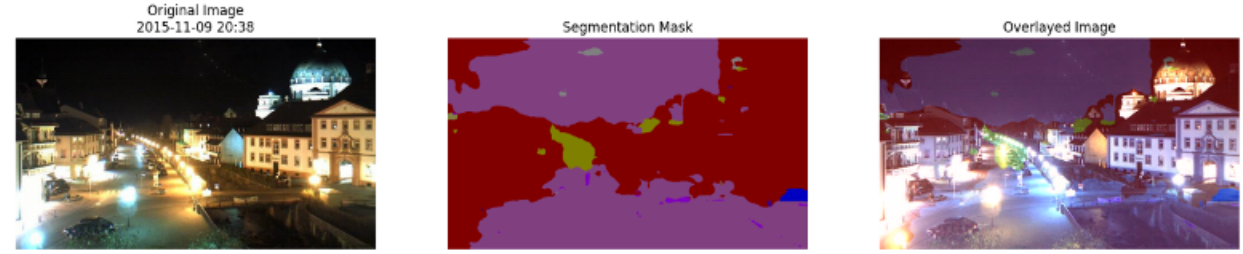

Figure 4: Segmentation Results with Overlay (Evening).

5.5 Segmentation Results with Overlay (Night)

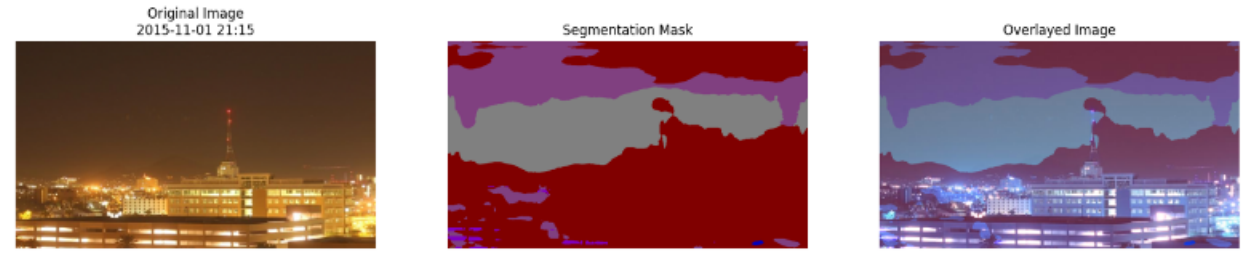

Figure 5: Segmentation Results with Overlay (Night).

5.6 Segmentation Results with Overlay after low-light enhancement and denoising (Night)

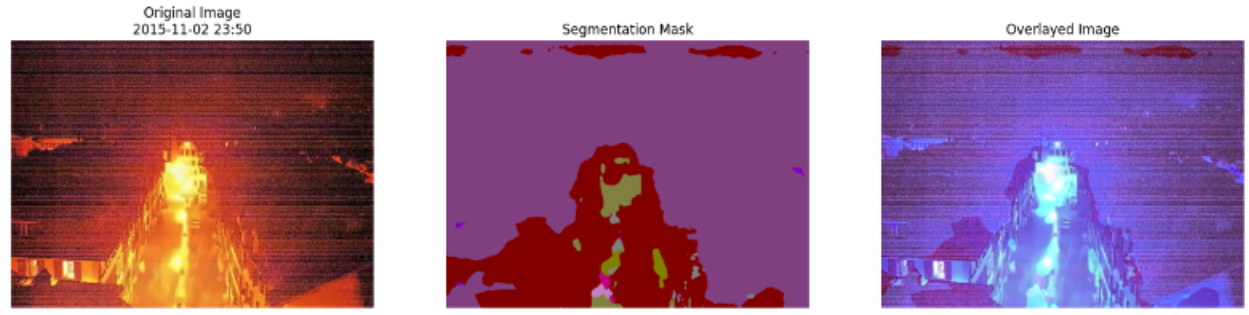

Figure 6: Segmentation Results with Overlay after low-light enhancement and denoising (Night).

6 Discussion

Baseline segmentation confidence is highest in morning conditions and lowest at night, consistent with expected illumination effects.

After low-light enhancement and denoising, night images show improved proxy metrics, including higher average confidence and more detected object classes.

Because the reported measures are proxy metrics (not IoU or mAP), improvements should be interpreted as changes in model output behavior rather than definitive accuracy gains.

7 Limitations and Future Work

- **No ground-truth masks**: Standard segmentation metrics (IoU, mIoU) cannot be computed.
- **Domain mismatch**: HRNet label space and training domain may not match the outdoor webcam scenes.
- **Sampling sensitivity**: If evaluation is performed on a subset of images, results may vary with sampling.

Future work includes running inference on all images per time category, exporting per-image metrics for reproducibility, and evaluating additional low-light enhancement methods.

References

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